

Experiment No. 10

Title:

Development of Social Media Text Analytics Models to Improve Existing Products/Services Using Customer Reviews

Aim:

To design and implement text analytics models that analyze customer reviews and comments from social media to identify insights for improving products or services.

Software Tools:

Google collab, Python programming.

Theory:

Social media platforms such as Twitter, Instagram, and Facebook contain large volumes of unstructured textual data in the form of comments, reviews, and posts.

Text Analytics

Text analytics refers to the process of converting unstructured text into meaningful insights using techniques from Natural Language Processing.

Sentiment Analysis

Classifies text into:

- Positive
- Negative
- Neutral

Text Preprocessing Steps:

- Tokenization
- Stop-word removal
- Stemming/Lemmatization
- Lowercasing

Feature Extraction Techniques:

- Bag of Words (BoW)
- TF-IDF (Term Frequency–Inverse Document Frequency)

Modeling Techniques:

- Machine Learning Models:
 - Naïve Bayes
 - Support Vector Machine (SVM)
- Deep Learning Models:
 - Recurrent Neural Networks (RNN)
 - LSTM

Topic Modeling

Used to identify common issues/themes in reviews (e.g., delivery delay, product quality). It automatically discover hidden themes (topics) from large volumes of unstructured social media text such as tweets, comments, and reviews.

Procedure:

<https://colab.research.google.com/drive/1AaYJTxhtmlTnMCVWHeROkhr-ORqxb0Xwh>

- Step 1: Data Collection
- Step 2: Data Preprocessing
- Step 3: Exploratory Data Analysis (EDA)
- Step 4: Feature Extraction
- Step 5: Model Development
- Step 6: Model Evaluation
- Step 7: Topic Modeling
- Step 8: Insight Generation



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

```
plt.rcParams["figure.figsize"] = (8,5)
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...  
[nltk_data] Unzipping tokenizers/punkt.zip.  
[nltk_data] Downloading package stopwords to /root/nltk_data...  
[nltk_data] Unzipping corpora/stopwords.zip.
```

```
data = {  
    "reviews": [  
        "The product quality is very good and delivery was fast",  
        "Customer service is very poor and response is slow",  
        "I love the design and packaging of the product",  
        "The product stopped working after one week",  
        "Great experience and affordable price",  
        "Bad customer support and late delivery",  
        "Excellent features and easy to use",  
        "Product quality is average but service is good",  
        "Very bad service and poor packaging",  
        "Amazing product and fast shipping",  
        "Worst product ever and support team not responding",  
        "Happy with product and delivery time"  
    ]  
}  
  
df = pd.DataFrame(data)  
df
```

reviews	
0	The product quality is very good and delivery ...
1	Customer service is very poor and response is ...
2	I love the design and packaging of the product
3	The product stopped working after one week
4	Great experience and affordable price
5	Bad customer support and late delivery
6	Excellent features and easy to use
7	Product quality is average but service is good



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

```
stop_words = set(stopwords.words('english'))

def preprocess(text):
    tokens = word_tokenize(text.lower())
    tokens = [word for word in tokens if word.isalpha()]
    tokens = [word for word in tokens if word not in stop_words]
    return " ".join(tokens)

df["clean_reviews"] = df["reviews"].apply(preprocess)

df
```

	reviews	clean_reviews
0	The product quality is very good and delivery ...	product quality good delivery fast
1	Customer service is very poor and response is ...	customer service poor response slow
2	I love the design and packaging of the product	love design packaging product
3	The product stopped working after one week	product stopped working one week
4	Great experience and affordable price	great experience affordable price
5	Bad customer support and late delivery	bad customer support late delivery
6	Excellent features and easy to use	excellent features easy use
7	Product quality is average but service is good	product quality average service good
8	Very bad service and poor packaging	bad service poor packaging
9	Amazing product and fast shipping	amazing product fast shipping
10	Worst product ever and support team not respon...	worst product ever support team responding
11	Happy with product and delivery time	happy product delivery time

Next steps: [Generate code with df](#) [New interactive sheet](#)



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

```
def get_sentiment(text):  
    analysis = TextBlob(text)  
    return analysis.sentiment.polarity  
  
df["sentiment_score"] = df["clean_reviews"].apply(get_sentiment)  
  
df["sentiment"] = df["sentiment_score"].apply(  
    lambda x: "Positive" if x > 0 else "Negative" if x < 0 else "Neutral"  
)  
  
df
```

reviews		clean_reviews	sentiment_score	sentiment
0	The product quality is very good and delivery ...	product quality good delivery fast	0.450000	Positive
1	Customer service is very poor and response is ...	customer service poor response slow	-0.350000	Negative
2	I love the design and packaging of the product	love design packaging product	0.500000	Positive
3	The product stopped working after one week	product stopped working one week	0.000000	Neutral
4	Great experience and affordable price	great experience affordable price	0.800000	Positive
5	Bad customer support and late delivery	bad customer support late delivery	-0.500000	Negative
6	Excellent features and easy to use	excellent features easy use	0.716667	Positive
7	Product quality is average but service is good	product quality average service good	0.275000	Positive
8	Very bad service and poor packaging	bad service poor packaging	-0.550000	Negative
9	Amazing product and fast shipping	amazing product fast shipping	0.400000	Positive
10	Worst product ever and support team not respon...	worst product ever support team responding	-1.000000	Negative
11	Happy with product and delivery time	happy product delivery time	0.800000	Positive

Next steps: [Generate code with df](#) [New interactive sheet](#)



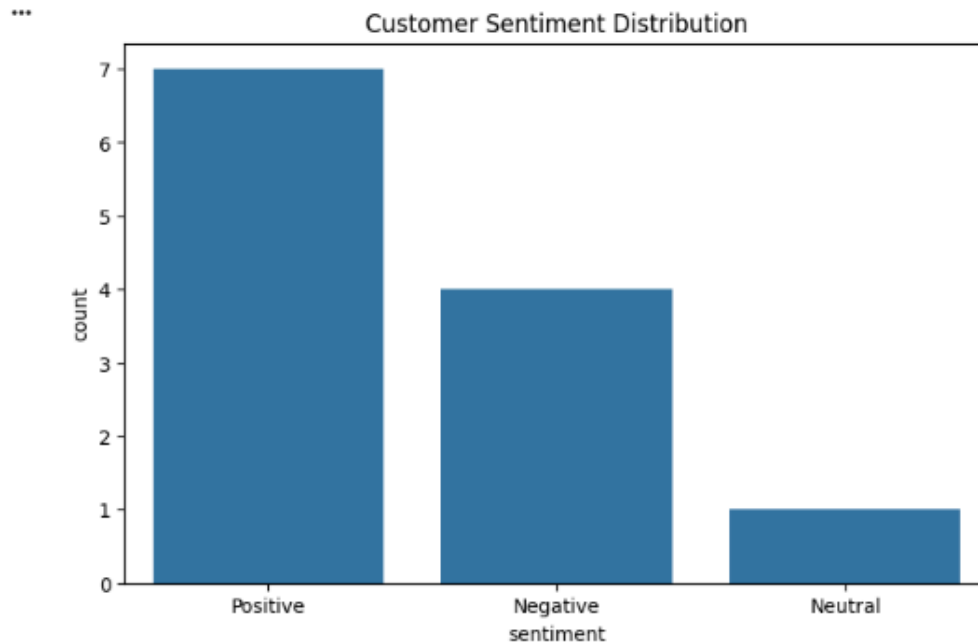
Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

```
sns.countplot(x="sentiment", data=df)  
plt.title("Customer Sentiment Distribution")  
plt.show()
```



s Terminal

Text analytics models help organizations understand customer opinions from social media data. By analyzing sentiments and extracting key topics, businesses can make data-driven decisions to improve their products and services.

Experiment No. 9

Title:

Analysis of Competitor Activities Using Social Media Data

Aim:

To analyze and evaluate competitor strategies, performance, and engagement patterns using data collected from social media platforms.

Software Tools:

Google collab, Python programming.

Theory:

Social media platforms such as Facebook, Instagram, Twitter, and LinkedIn generate large amounts of user interaction data. This data can be analyzed to understand competitor behavior.

Competitor Analysis

The process of identifying strengths, weaknesses, strategies, and performance of competing brands. It enables businesses to discover market opportunities, mitigate risks, and build a competitive advantage by analyzing customer feedback, pricing, and content strategies. Competitor analytics helps to identify competitors and research their marketing strategies. The key metrics involved in Competitor analytics include:

- Engagement Rate
- Follower Growth
- Content Reach
- Posting Frequency
- Hashtag Usage

Types of Analysis:

- Content Analysis: Type of posts (videos, images, reels, text)
- Sentiment Analysis: Positive/negative audience reactions
- Trend Analysis: Popular topics and hashtags
- Benchmarking: Comparing metrics across competitors



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

Procedure:

https://colab.research.google.com/drive/1EKHa3XKcNcftDtfKgaS6DHG_mszegmOR

1. Generate sample dataset
2. Perform data cleaning
3. Analyze engagement trends
4. Visualize competitor performance
5. Perform sentiment analysis



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from textblob import TextBlob
```

Step 1: Generate Sample Social Media Data



```
np.random.seed(42)

brands = ['BrandA', 'BrandB', 'BrandC']
dates = pd.date_range(start='2025-01-01', periods=100)

data = []

for date in dates:
    for brand in brands:
        post = f"{brand} launches new product"
        likes = np.random.randint(100, 1000)
        comments = np.random.randint(10, 200)
        shares = np.random.randint(5, 100)

        data.append([date, brand, post, likes, comments, shares])

df = pd.DataFrame(data, columns=['Date', 'Brand', 'Post', 'Likes', 'Comments', 'Shares'])
df.head()
```

	Date	Brand	Post	Likes	Comments	Shares
0	2025-01-01	BrandA	BrandA launches new product	202	189	97
1	2025-01-01	BrandB	BrandB launches new product	370	116	76
2	2025-01-01	BrandC	BrandC launches new product	800	30	87
3	2025-01-02	BrandA	BrandA launches new product	314	84	79



Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

Step 2: Calculate Engagement



```
df['Engagement'] = df['Likes'] + df['Comments'] + df['Shares']  
df.head()
```

	Date	Brand	Post	Likes	Comments	Shares	Engagement	Sentiment
0	2025-01-01	BrandA	BrandA launches new product	202	189	97	488	0.136364
1	2025-01-01	BrandB	BrandB launches new product	370	116	76	562	0.136364
2	2025-01-01	BrandC	BrandC launches new product	800	30	87	917	0.136364
3	2025-01-02	BrandA	BrandA launches new product	314	84	79	477	0.136364
4	2025-01-02	BrandB	BrandB launches new product	187	126	28	341	0.136364

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Step 3: Competitor Performance Analysis

```
performance = df.groupby('Brand')['Engagement'].mean().reset_index()  
performance
```

	Brand	Engagement
0	BrandA	696.61
1	BrandB	706.93
2	BrandC	695.14

Next steps:

[Generate code with performance](#)[New interactive sheet](#)



Shree Rahul Education Society's (Regd.)

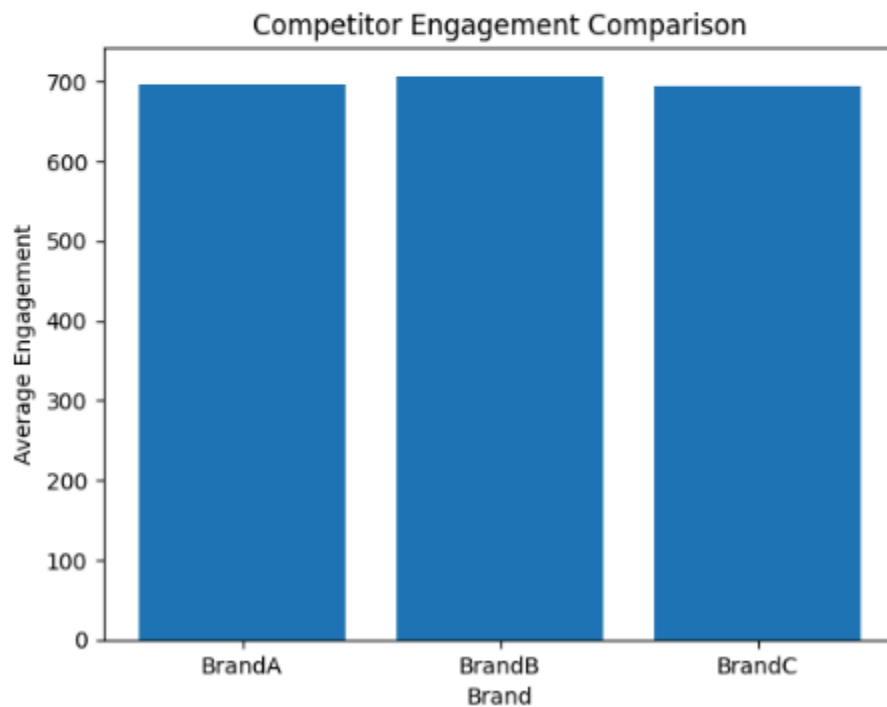
SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

Step 4: Visualization

```
plt.figure()
plt.bar(performance['Brand'], performance['Engagement'])
plt.xlabel('Brand')
plt.ylabel('Average Engagement')
plt.title('Competitor Engagement Comparison')
plt.show()
```





Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

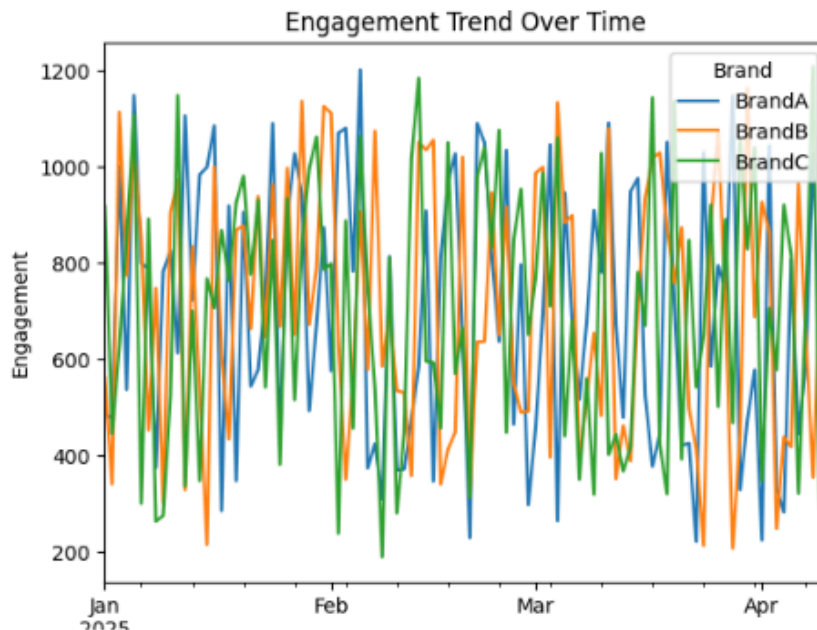
Step 5: Engagement Trend Over Time



```
trend = df.groupby(['Date', 'Brand'])['Engagement'].mean().unstack()

plt.figure()
trend.plot()
plt.title('Engagement Trend Over Time')
plt.xlabel('Date')
plt.ylabel('Engagement')
plt.show()
```

*** <Figure size 640x480 with 0 Axes>





Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UOC Act 1956 | Minority Status (Hindi Linguistic)

Step 6: Sentiment Analysis



```
def get_sentiment(text):  
    return TextBlob(text).sentiment.polarity  
  
df['Sentiment'] = df['Post'].apply(get_sentiment)  
  
sentiment_summary = df.groupby('Brand')['Sentiment'].mean().reset_index()  
sentiment_summary
```

	Brand	Sentiment
0	BrandA	0.136364
1	BrandB	0.136364
2	BrandC	0.136364

Next steps:

[Generate code with sentiment_summary](#)

[New interactive sheet](#)

Conclusion:

Social media data provides valuable insights into competitor strategies. By analyzing engagement patterns and content types, businesses can refine their own marketing strategies to improve performance and audience interaction.



Experiment No. 8

TITLE: Social Media Engagement Prediction using Data Analysis and Visualization Techniques

AIM: To analyze social media post data and predict engagement using machine learning techniques and visualize actual vs predicted engagement using bar graphs.

THEORY:

1. Introduction to Social Media Engagement Prediction

Social media analytics helps measure user interaction such as likes, shares and comments. These metrics are combined to form an engagement score. Machine learning algorithms like Linear Regression can be used to identify patterns in data and predict engagement based on platform, content type and posting time. Data preprocessing is performed using encoding techniques to convert categorical values into numerical form. The dataset is divided into training and testing sets to build and evaluate the model. The predicted engagement is compared with actual engagement using graphical visualization.

Social Media Analytics focuses on extracting useful insights from user interactions such as likes, shares, and comments. These interactions indicate how users respond to different types of content. In this experiment, machine learning is used to predict engagement based on platform, content type, and posting time. Predicting engagement helps in planning better social media strategies and improving content performance.

The experiment uses data visualization and regression analysis to compare actual and predicted engagement values. This approach helps businesses and individuals identify which type of content performs better and when it should be posted.

2. Fundamental Concepts in Social Media Data Analysis

2.1 Social Media Dataset Representation

Social media data is organized in tabular format using DataFrame. Each row represents one post and contains attributes such as:

- Platform (Instagram, Facebook)
- Content Type (Reel, Post, Video, Story)
- Posting Time (Morning, Evening, Night)
- Likes
- Shares
- Comments

This structured representation helps in performing analysis efficiently.

2.2 Engagement Metric

Engagement measures user interaction with content. Engagement Score is calculated using the formula:

$$\text{Engagement} = \text{Likes} + \text{Shares} + \text{Comments}$$

This metric helps identify:

- High-performing content
- User interaction level
- Effectiveness of posting strategy

2.3 Data Preprocessing

Machine learning models require numerical data. Since the dataset contains categorical values such as platform and content type, one-hot encoding is used to convert them into numerical format. This process creates binary columns representing each category and prepares data for model training.

3. Core Techniques in Engagement Prediction

3.1 Train-Test Split

The dataset is divided into training and testing sets. Training data is used to build the model and testing data is used to evaluate performance. This ensures that the model can generalize to new data.

3.2 Linear Regression Model

Linear Regression is a supervised machine learning algorithm used to predict continuous values. It establishes a relationship between independent variables (platform, content type, posting time) and dependent variable (engagement). The model learns coefficients that indicate the influence of each feature on engagement.

3.3 Prediction of Engagement

After training, the model predicts engagement values for the test dataset. These predicted values are compared with actual engagement values to evaluate performance.

3.4 Data Visualization

Bar charts are used to visualize:

- Actual engagement values
- Predicted engagement values

Visualization helps in comparing model predictions with real data and understanding performance.

□ Practical 8: Social Media Engagement Prediction

□ Step 1: Install Required Libraries

Install required libraries for data analysis and machine learning.

```
!pip install pandas numpy matplotlib scikit-learn
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (fro
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from
```

□ Step 2: Import Libraries

Import all necessary Python libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
```

□ Step 3: Create Social Media Dataset

```
data = {
    'Platform': ['Instagram','Instagram','Facebook','Facebook','Instagram','Facebook','Inst
    'Content_Type': ['Reel','Post','Video','Post','Story','Video','Reel','Story'],
    'Posting_Time': ['Morning','Evening','Night','Morning','Night','Evening','Evening','Nig
    'Likes': [120,80,60,40,100,55,130,70],
    'Shares': [30,10,20,5,25,15,35,10],
    'Comments': [20,5,10,3,18,8,22,7]
}
```



```
df = pd.DataFrame(data)
df
```

	Platform	Content_Type	Posting_Time	Likes	Shares	Comments	
0	Instagram	Reel	Morning	120	30	20	
1	Instagram	Post	Evening	80	10	5	
2	Facebook	Video	Night	60	20	10	
3	Facebook	Post	Morning	40	5	3	
4	Instagram	Story	Night	100	25	18	
5	Facebook	Video	Evening	55	15	8	
6	Instagram	Reel	Evening	130	35	22	
7	Facebook	Story	Night	70	10	7	

Next steps: [Generate code with df](#) [New interactive sheet](#)

☐ Step 4: Convert Data into DataFrame

```
print(df)
```

	Platform	Content_Type	Posting_Time	Likes	Shares	Comments
0	Instagram	Reel	Morning	120	30	20
1	Instagram	Post	Evening	80	10	5
2	Facebook	Video	Night	60	20	10
3	Facebook	Post	Morning	40	5	3
4	Instagram	Story	Night	100	25	18
5	Facebook	Video	Evening	55	15	8
6	Instagram	Reel	Evening	130	35	22
7	Facebook	Story	Night	70	10	7

☐ Step 5: Convert Text Data into Numbers

```
df_encoded = pd.get_dummies(df, columns=['Platform','Content_Type','Posting_Time'])
```

☐ Step 6: Create Engagement Score

```
df_encoded['Engagement'] = df['Likes'] + df['Shares'] + df['Comments']
```

☐ Step 7: Define Input and Output

```
X = df_encoded.drop(['Likes','Shares','Comments','Engagement'], axis=1) y  
= df_encoded['Engagement']
```

☐ Step 8: Split Data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=4
```

□ Step 9: Create Machine Learning Model

```
model = LinearRegression() model.fit(X_train, y_train)
```

▼ **LinearRegression** ⓘ ?
LinearRegression()

□ Step 10: Predict Engagement

```
y_pred = model.predict(X_test)
```

□ Step 11: Check Error

```
mse = mean_squared_error(y_test, y_pred)  
print('Mean Squared Error:', mse)
```

Mean Squared Error: 560.6050000000001

□ Step 12: Feature Importance

```
importance = pd.DataFrame({  
    'Feature': X.columns,  
    'Coefficient': model.coef_  
})  
importance
```

	Feature	Coefficient	
0	Platform_Facebook	-28.0	 
1	Platform_Instagram	28.0	
2	Content_Type_Post	-32.8	
3	Content_Type_Reel	33.2	
4	Content_Type_Story	-1.7	
5	Content_Type_Video	1.3	
6	Posting_Time_Evening	8.7	
7	Posting_Time_Morning	-8.3	
8	Posting_Time_Night	-0.4	

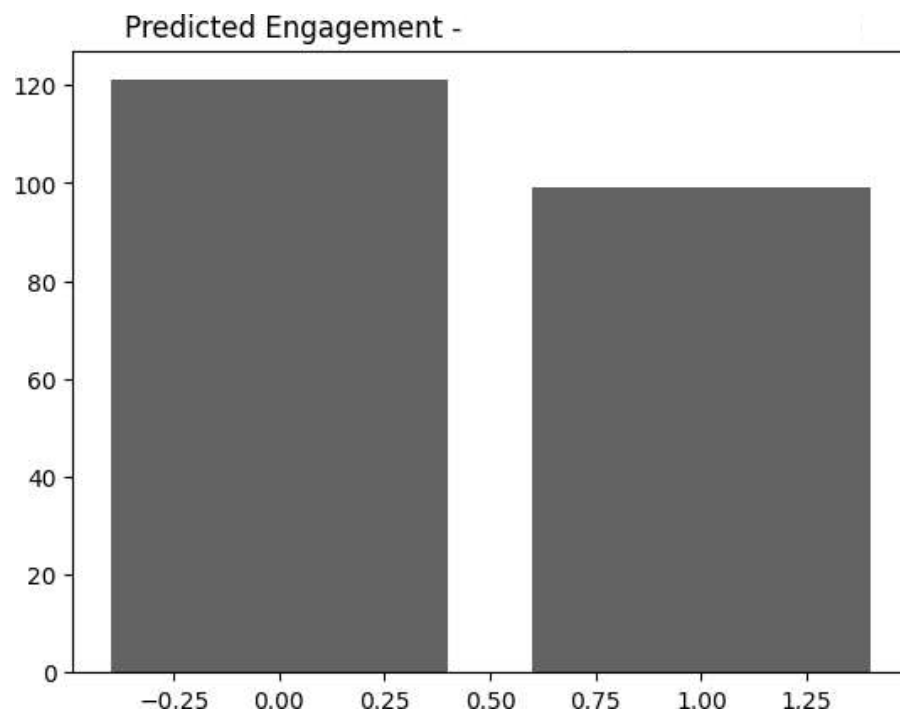
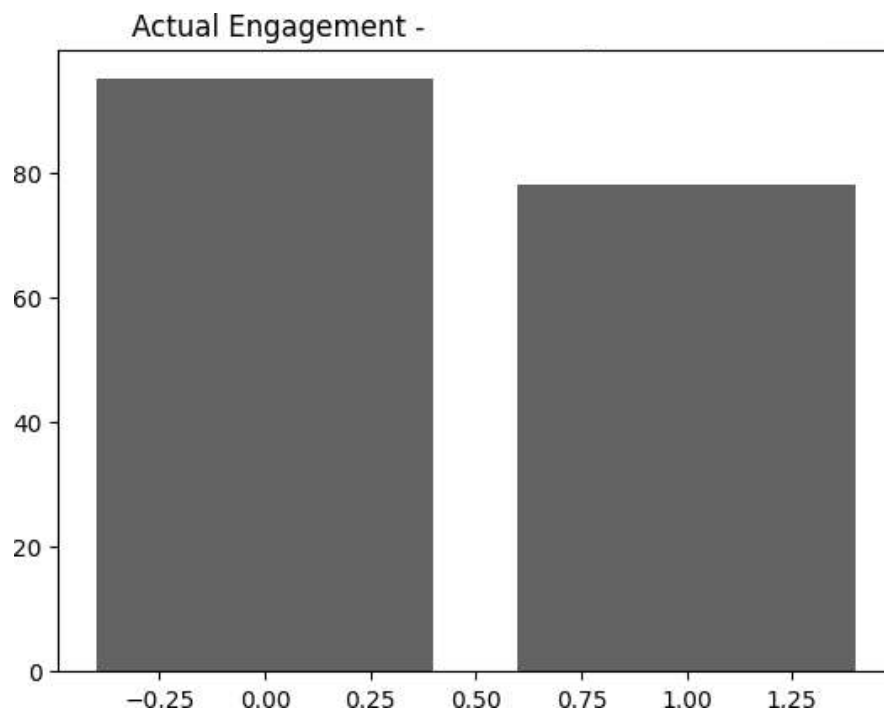
Next steps: [Generate code with importance](#)

[New interactive sheet](#)

□ Step 13: Plot Graph

```
plt.figure()  
plt.bar(range(len(y_test)), y_test)  
plt.title('Actual Engagement - ( Leenal Rajendra Bura- 06)') plt.show()
```

```
plt.figure()  
plt.bar(range(len(y_pred)), y_pred)  
plt.title('Predicted Engagement - ( Leenal Rajendra Bura- 06)')  
plt.show()
```



□ Step 14: Conclusion

```
print('Reels and Videos tend to give higher engagement.')  
print('Evening and Night posting shows better performance.')  
print('Instagram performs better than Facebook.')
```

Reels and Videos tend to give higher engagement.
Evening and Night posting shows better performance.
Instagram performs better than Facebook.



Experiment No. 7

TITLE: Social Media Analytics Dashboard using Data Visualization Techniques

AIM: To design and develop a social media analytics dashboard using data visualization techniques to analyze user engagement, sentiment distribution, and trending topics.

THEORY:

1. Introduction to Social Media Analytics Dashboard

Social Media Analytics focuses on extracting meaningful insights from user-generated data on social platforms. Unlike structure-based analytics, which studies relationships between users, this experiment emphasizes data-driven insights using engagement metrics and visualization.

A dashboard provides a visual interface to monitor:

- User engagement (likes, comments, shares)
- Sentiment trends (positive, negative, neutral)
- Trending topics (hashtags)
- High-performing content

Such dashboards are widely used by businesses for marketing strategies, audience analysis, and decision-making.

2. Fundamental Concepts in Social Media Data Analysis

2.1 Data Representation

Social media data is structured into tabular format using DataFrames.

Each row represents a post with attributes such as:

- User
- Post content
- Likes, comments, shares
- Hashtag
- Sentiment

This structured representation helps in efficient analysis.

2.2 Engagement Metrics

Engagement measures how users interact with content.

Engagement Score Formula:

$$\text{Engagement} = \text{Likes} + \text{Comments} + \text{Shares}$$

This metric helps identify:



- Popular posts
- User interaction levels
- Content effectiveness

2.3 Data Simulation

Since real-time API data is not used, synthetic data is generated using tools like Faker.

Generated data includes:

- Random usernames
- Random posts
- Random engagement values
- Random sentiment labels

This simulates real-world social media activity.

3. Core Techniques in Social Media Analytics

3.1 Sentiment Analysis

Sentiment analysis classifies posts into:

- Positive
- Negative
- Neutral

It helps understand public opinion and user mood.

Visualization:

- Pie chart showing sentiment distribution

Business Application:

- Brand perception analysis
- Customer feedback monitoring

3.2 Trending Hashtag Analysis

Hashtags represent topics or trends in social media.

Frequency analysis is used to identify:

- Most used hashtags
- Popular topics

Visualization:

- Bar chart of hashtag counts



Business Application:

- Trend identification
- Campaign optimization

3.3 Engagement Analysis

Analyzes relationship between:

- Likes
- Shares
- Sentiment

Visualization:

- Scatter plot
- Bubble size represents engagement

Business Application:

- Identify high-performing content
- Optimize post strategies

3.4 Viral Content Identification

Posts with highest engagement are considered viral.

Technique:

- Sorting posts by engagement score
- Selecting top-performing posts

Business Application:

- Content strategy improvement
- Influencer collaboration

3.5 Report Generation

A summary report provides key statistics:

- Total posts
- Average likes, comments, shares
- Most trending hashtag

Business Application:

- Quick insights for decision-making
- Performance tracking

4. Applications in Business

Social Media Analytics Dashboard is used in:

- Marketing Analysis: Understand audience behavior
- Brand Monitoring: Track sentiment and reputation
- Content Strategy: Identify what works best
- Trend Analysis: Detect trending topics
- Customer Engagement: Improve interaction strategies

5. Steps to Implement Social Media Analytics Dashboard Model

1. Open Google Colab and create a new notebook

 `!pip install pandas plotly faker`

```
... Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Collecting faker
  Downloading faker-40.11.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from plotly) (26.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading faker-40.11.0-py3-none-any.whl (2.0 MB)
----- 2.0/2.0 MB 22.1 MB/s eta 0:00:00
Installing collected packages: faker
Successfully installed faker-40.11.0
```

2. Install required libraries (pandas, plotly, faker)

[2]  `import pandas as pd`
`import random`
`import plotly.express as px`
`from faker import Faker`

3. Generate simulated social media dataset

```
fake = Faker()

sentiments = ['Positive', 'Negative', 'Neutral']
hashtags = ['#AI', '#Tech', '#DataScience', '#Python', '#Startup']

data = []

for i in range(200):
    post = {
        "user": fake.user_name(),
        "post_text": fake.sentence(),
        "sentiment": random.choice(sentiments),
        "likes": random.randint(10,500),
        "comments": random.randint(1,100),
        "shares": random.randint(1,50),
        "hashtag": random.choice(hashtags)
    }
    data.append(post)

df = pd.DataFrame(data)

df.head()
```

	user	post_text	sentiment	likes	comments	shares	hashtag
0	tracycameron	Speak bank suffer result prevent.	Neutral	178	100	4	#Tech
1	pbrown	Someone generation study case nor.	Positive	295	71	31	#Startup
2	lowen	Since notice character much else business share.	Neutral	14	29	19	#DataScience
3	eayers	Central live standard water.	Positive	231	11	6	#Tech
4	christopher23	Minute mission number lot or against maintain ...	Neutral	474	39	37	#Python



4. Create engagement metric

```
df["engagement"] = df["likes"] + df["comments"] + df["shares"]  
df.head()
```

	user	post_text	sentiment	likes	comments	shares	hashtag	engagement
0	jamesmccullough	Institution beautiful each culture mother.	Positive	376	60	45	#Startup	481
1	elizabeth03	Go purpose want big question color.	Neutral	417	1	45	#Startup	463
2	jsingleton	Success it group popular teacher member.	Neutral	290	11	10	#DataScience	311
3	jparsons	Technology wide learn peace time hot set.	Neutral	311	95	4	#Tech	410
4	james63	Guess fund pay strategy.	Neutral	151	100	27	#Startup	278

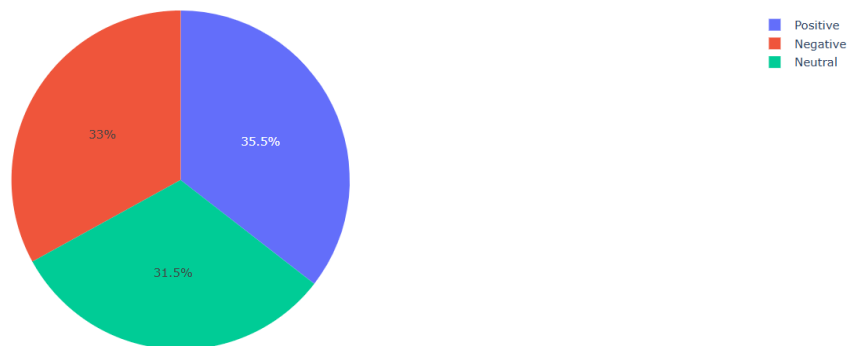
5. Perform sentiment analysis and visualize results

```
sentiment_count = df["sentiment"].value_counts().reset_index()  
sentiment_count.columns = ["Sentiment", "Count"]
```

```
fig = px.pie(  
    sentiment_count,  
    values="Count",  
    names="Sentiment",  
    title="Sentiment Distribution"  
)
```

```
fig.show()
```

Sentiment Distribution



6. Analyze trending hashtags using bar charts

```
hashtag_count = df["hashtag"].value_counts().reset_index()  
hashtag_count.columns = ["Hashtag", "Count"]
```

```
fig = px.bar(  
    hashtag_count,  
    x="Hashtag",  
    y="Count",  
    title="Trending Hashtags"  
)
```

```
fig.show()
```



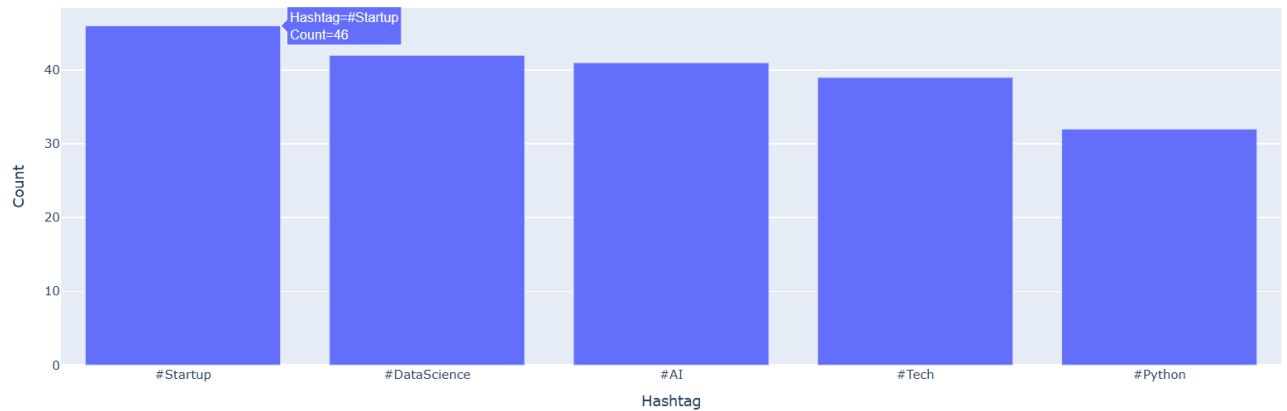
Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified |
DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

Trending Hashtags



7. Perform engagement analysis using scatter plots

```
fig = px.scatter(  
    df,  
    x="likes",  
    y="shares",  
    color="sentiment",  
    size="engagement",  
    title="Engagement Analysis"  
)
```

```
fig.show()
```

Engagement Analysis





8. Identify top viral posts

```
top_posts = df.sort_values(by="engagement", ascending=False).head(10)
```

```
top_posts[["user", "post_text", "engagement"]]
```

...	user	post_text	engagement
151	mboyd	Technology past evening rich.	626
143	frederickjames	Believe professor us student appear similar.	623
13	williamsbrandon	Maybe simply thousand least number.	596
168	colemanbeth	How wait course of hold.	596
63	smithtracy	Measure career produce move matter he news nice.	587
123	ijames	Make book at year set industry.	585
94	richardburke	Lay success voice use the themselves.	580
24	christopher35	Work song relationship financial see suggest h...	575
163	christinehamilton	Knowledge keep position fund add hundred vote ...	574
149	waltersrussell	Various business court job.	569

9. Generate a summary report

```
report = {  
    "Total Posts": len(df),  
    "Average Likes": df["likes"].mean(),  
    "Average Comments": df["comments"].mean(),  
    "Average Shares": df["shares"].mean(),  
    "Most Trending Hashtag": df["hashtag"].mode()[0]  
}
```

```
report
```

```
... {'Total Posts': 200,  
     'Average Likes': np.float64(267.3),  
     'Average Comments': np.float64(51.735),  
     'Average Shares': np.float64(25.315),  
     'Most Trending Hashtag': '#Startup'}
```

6. Conclusion

The experiment demonstrates how a Social Media Analytics Dashboard can be developed using data visualization techniques. It helps in analyzing engagement, identifying trends, and understanding user sentiment. Such dashboards play a crucial role in modern business environments for data-driven decision-making and digital marketing strategies.